CRM APPLICATION FOR THE TEA SHOP & CO

PROJECT DOCUMENTATION

A Customer Relationship Management System for managing customers, products, orders, and business activities

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Dev Environment: Visual Studio Code
Database: MySQL

CERTIFICATE

This is to certify that the project entitled **"CRM Application for The Tea Shop & Co"** is a project work prepared as part of the academic/internship project requirements. The system is designed to demonstrate the use of web technologies, Python-based backend development, and MySQL database management for a small business CRM application.

Place: _____ Signature of Guide: _____

Date: _____ Signature of Student: _____

DECLARATION

I hereby declare that the project titled **"CRM Application for The Tea Shop & Co"** is prepared for academic/project purposes. The project demonstrates the design and development of a customer relationship management application using a web-based interface, Python backend, and MySQL database.

Acknowledgement

I express my sincere thanks to my project guide and everyone who supported me in completing this project. I am grateful for the guidance received during requirement analysis, database design, programming, testing, and documentation. I also thank my institution, friends, and family for their encouragement and support.

ABSTRACT

The CRM Application for The Tea Shop & Co is a web-based business management system developed to organize customer information and day-to-day shop activities in a centralized database. The application provides a simple interface for managing customer records, products, orders, and related business information. The system uses a Python web backend, a browser-based frontend, and MySQL as the relational database. The main purpose is to reduce manual record keeping, improve data organization, and make customer and order information easier to access. The project is suitable as a small-business CRM foundation and can be extended with billing, reports, authentication, notifications, and analytics.

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1. INTRODUCTION

Customer Relationship Management (CRM) is a method of organizing customer-related information and interactions so that a business can provide better service and maintain accurate records. For a tea shop or small food business, customer details, product information, and order records can become difficult to manage when they are maintained manually. This project proposes a simple web-based CRM application that keeps these records in a MySQL database.

The application is intended to be easy to operate and suitable for development and maintenance using Visual Studio Code. The frontend provides forms and pages for users, while the Python backend processes requests and communicates with the database.

2. PROBLEM STATEMENT

- Manual customer records can be difficult to search and update.
- Order information may be stored in separate notebooks or files.
- Repeated data entry can cause errors and duplicate records.
- Business information is not available in one centralized location.
- Generating useful customer and sales information manually takes time.

3. OBJECTIVES

- Create a simple CRM application for The Tea Shop & Co.
- Store customer details in a centralized MySQL database.
- Manage products and their basic information.
- Record and manage customer orders.
- Provide search, add, edit, and delete operations where appropriate.
- Reduce manual paperwork and improve data accuracy.

4. EXISTING & PROPOSED SYSTEM

Existing System: In a basic manual system, customer and order information is maintained in notebooks or spreadsheets. This leads to difficult searching, high risk of duplicate entries, limited analytics, and lack of central access.

Proposed System: A browser-based CRM connected to a MySQL database via Python. It centralizes customer, product, and order management with full CRUD capability and validated inputs.

5. SYSTEM REQUIREMENTS

Hardware Requirements:

- 4 GB RAM minimum
- Dual-core processor or better
- 5 GB storage space

Software Requirements:

- Windows / Linux / macOS
- VS Code IDE & Python 3.x
- MySQL Server & Workbench

6. TECHNOLOGIES USED

Technology	Purpose	Use in Project
HTML5	Page structure	Forms, tables, and application pages
CSS3	Presentation	Layout, spacing, colors, responsive design
JavaScript	Client behavior	Validation, dynamic UI interactions
Python 3.x	Backend logic	Application routing, business logic
Flask	Web Framework	Routes, REST APIs, database integration
MySQL	Database	Persistent relational storage

7. ARCHITECTURE & MODULES

System Architecture: 3-tier architecture (Presentation Layer → Application Layer → Data Layer).

Modules: Dashboard, Customer Management, Product Management, Order Management, Database Operations, Input Validation.

8. DATABASE DESIGN

Table	Primary Key	Key Fields	Purpose
customers	customer_id	name, phone, email, address	Customer master records
products	product_id	name, category, price, stock	Shop product catalog
orders	order_id	customer_id, order_date, total	Order transaction header
order_items	order_item_id	order_id, product_id, quantity	Itemized order breakdown

9. IMPLEMENTATION SQL & BACKEND CODE

```
CREATE DATABASE tea_shop_crm; USE tea_shop_crm; CREATE TABLE customers ( customer_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100) NOT NULL, phone VARCHAR(20), email VARCHAR(100), address VARCHAR(255), created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP ); CREATE TABLE products ( product_id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100) NOT NULL, category VARCHAR(50), price DECIMAL(10,2) NOT NULL, stock INT DEFAULT 0 );
```

```
# Flask Backend Connection Sample (app.py) from flask import Flask import mysql.connector app = Flask(__name__) db = mysql.connector.connect( host="localhost", user="root", password="YOUR_PASSWORD", database="tea_shop_crm" ) @app.route("/") def home(): return "Tea Shop CRM is running successfully" if __name__ == "__main__": app.run(debug=True)
```

10. SYSTEM SCREENSHOTS & APPLICATION INTERFACES

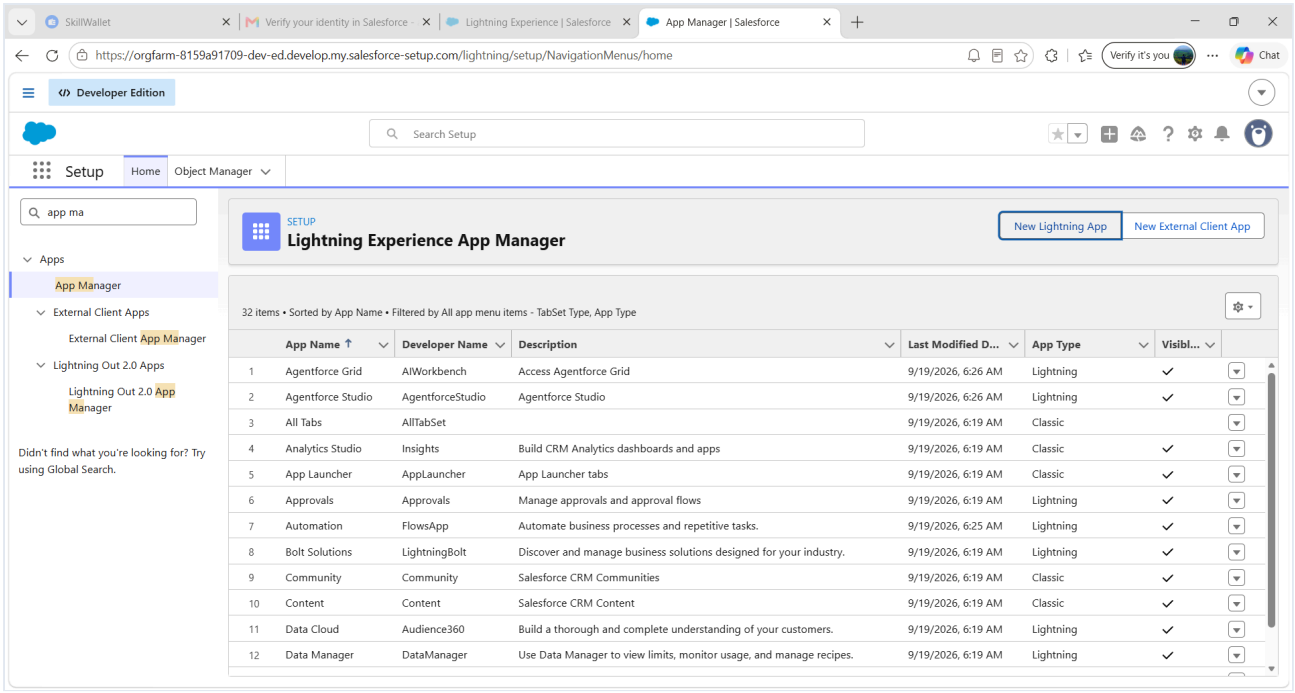


Figure 1: Salesforce App Manager & Navigation Menu Configuration

Configuring the primary application structure and navigation settings for business modules.

SYSTEM SCREENSHOTS (CONTINUED)

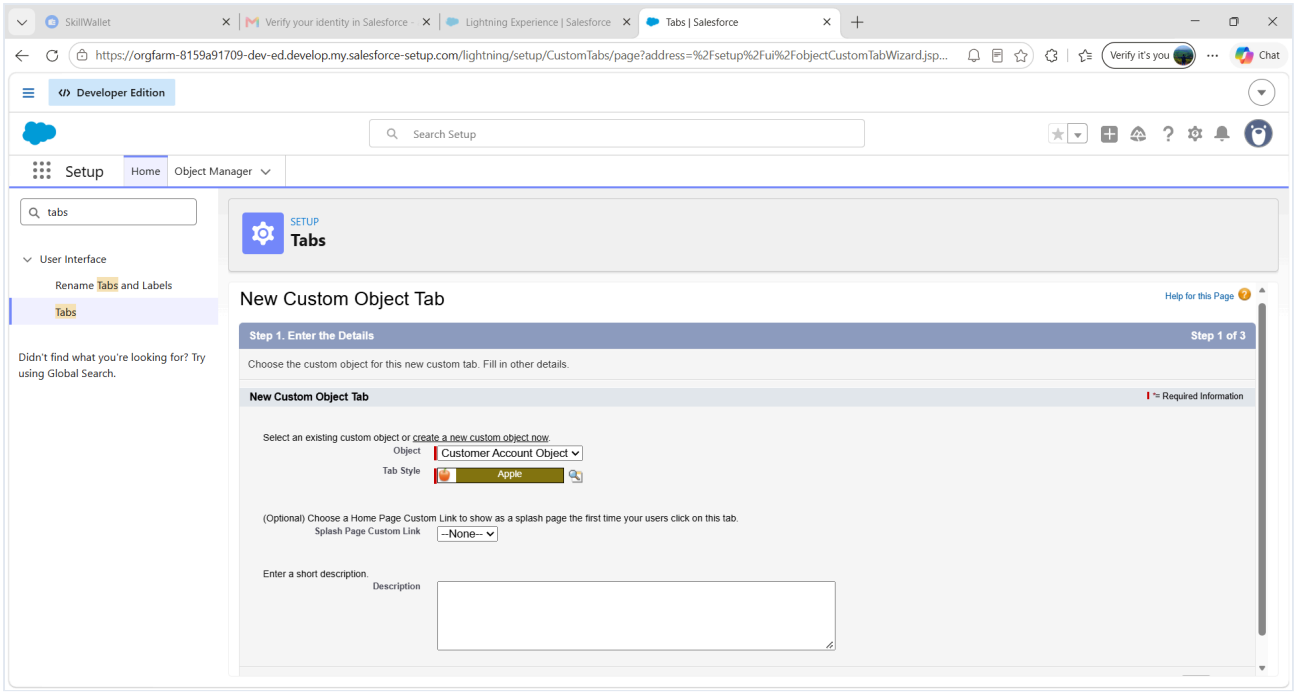


Figure 2: Custom Tab Setup for Customer Account Object

Setting up custom navigation tabs and visual style definitions for managing customer accounts.

SYSTEM SCREENSHOTS (CONTINUED)

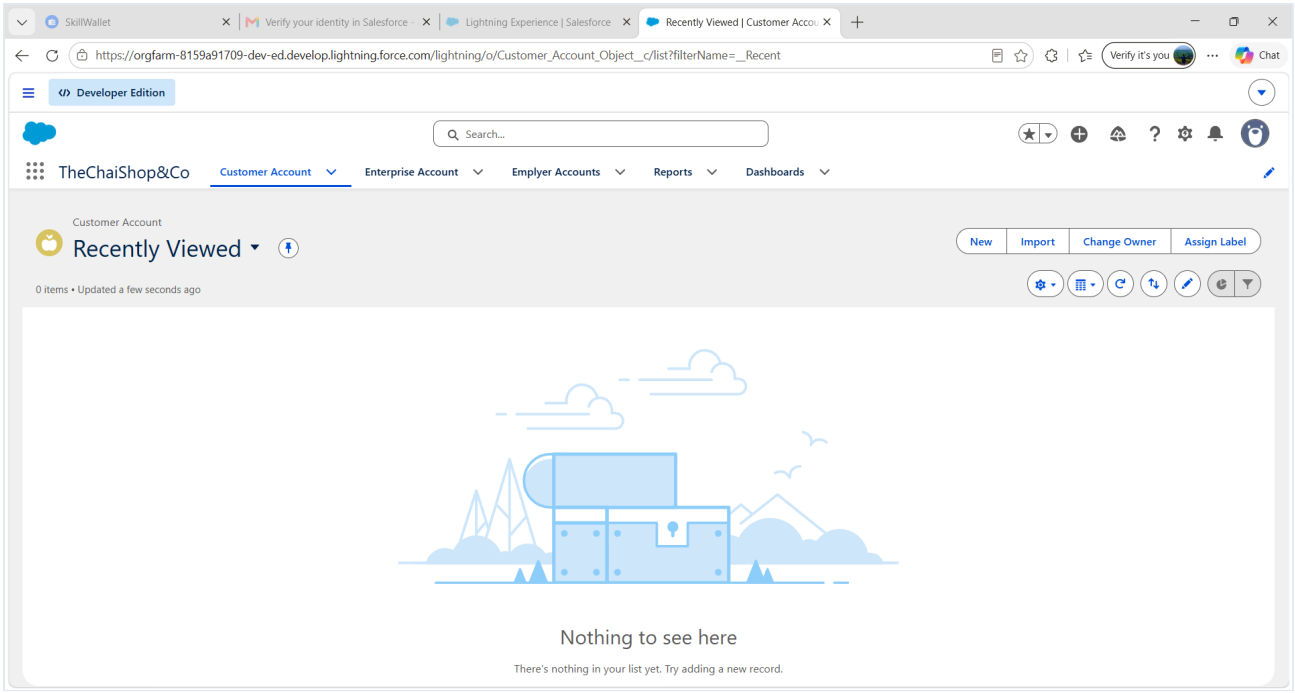


Figure 3: Customer Account List View Dashboard

Main customer account management view for quick searching, filtering, and record management.

SYSTEM SCREENSHOTS (CONTINUED)

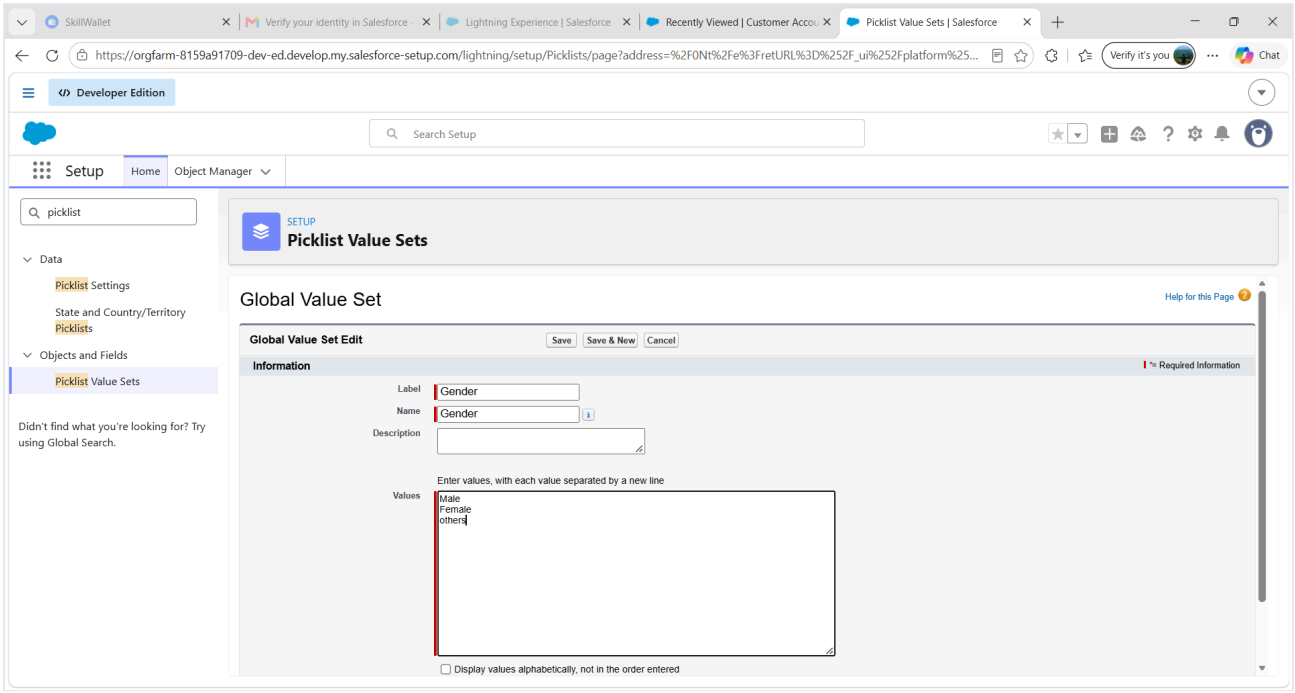


Figure 4: Global Picklist Configuration (Gender Values)

Defining reusable value sets and field dropdown items for consistent customer data entry.

SYSTEM SCREENSHOTS (CONTINUED)

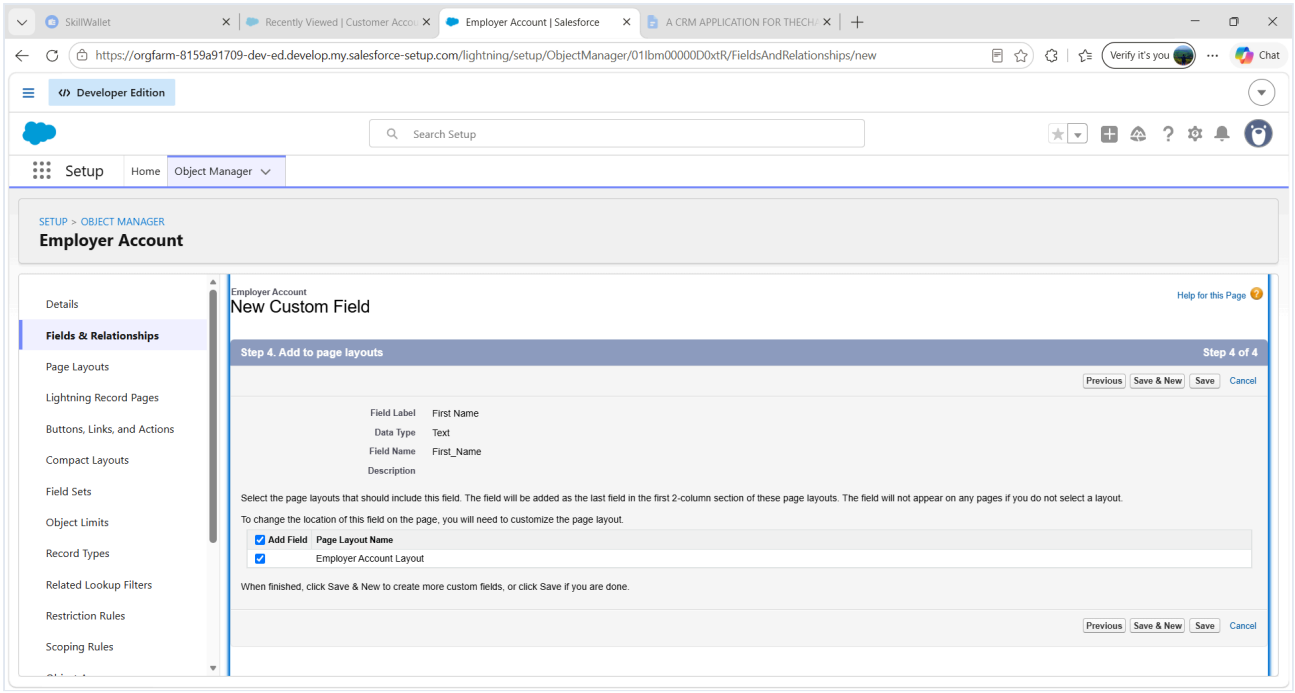


Figure 5: Adding Custom Field - First Name

Field schema step mapping new text fields to page layouts across customer objects.

SYSTEM SCREENSHOTS (CONTINUED)

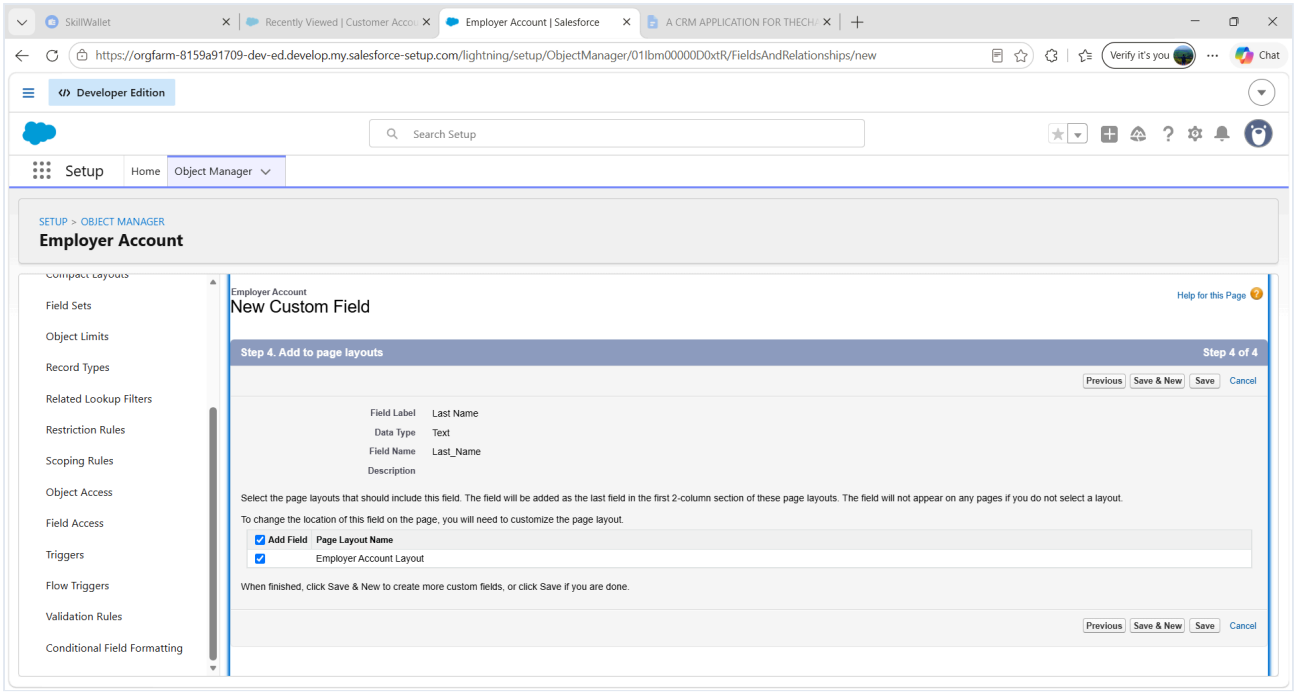


Figure 6: Adding Custom Field - Last Name

Assigning custom fields to specified layout sections for streamlined record entry.

SYSTEM SCREENSHOTS (CONTINUED)

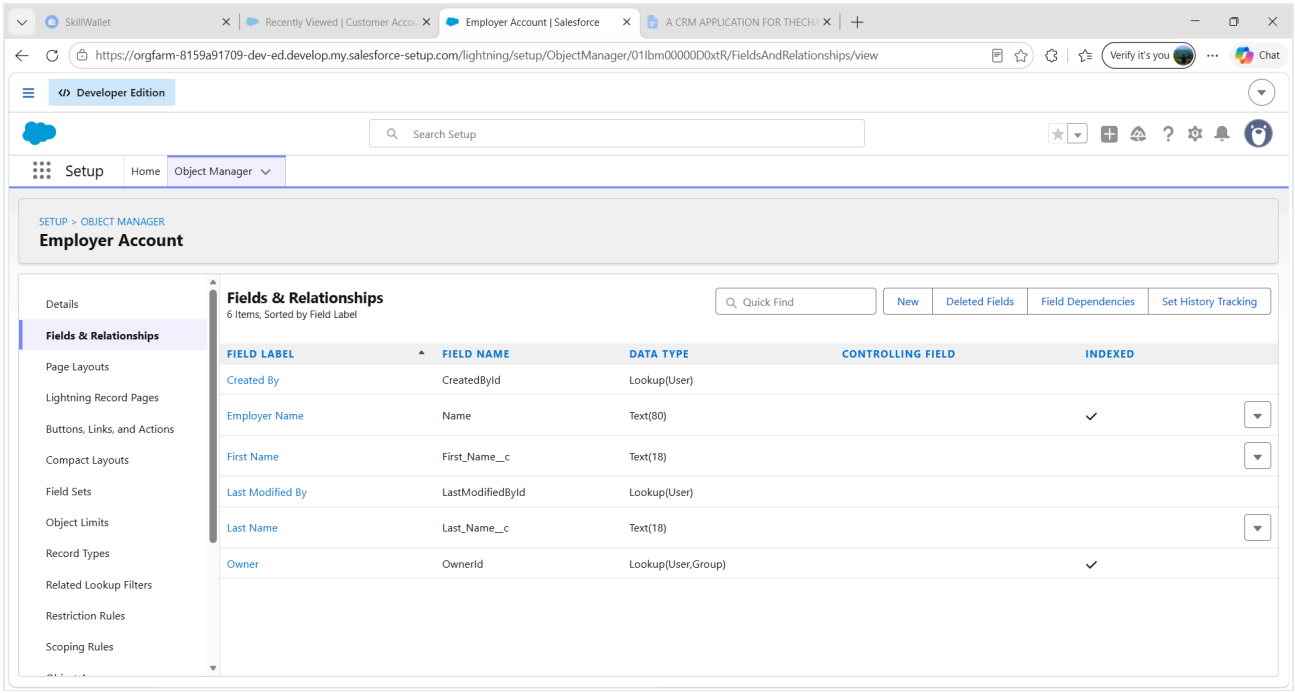


Figure 7: Fields & Relationships Summary Matrix

Comprehensive summary of all standard and custom attributes associated with Employer Accounts.

SYSTEM SCREENSHOTS (CONTINUED)

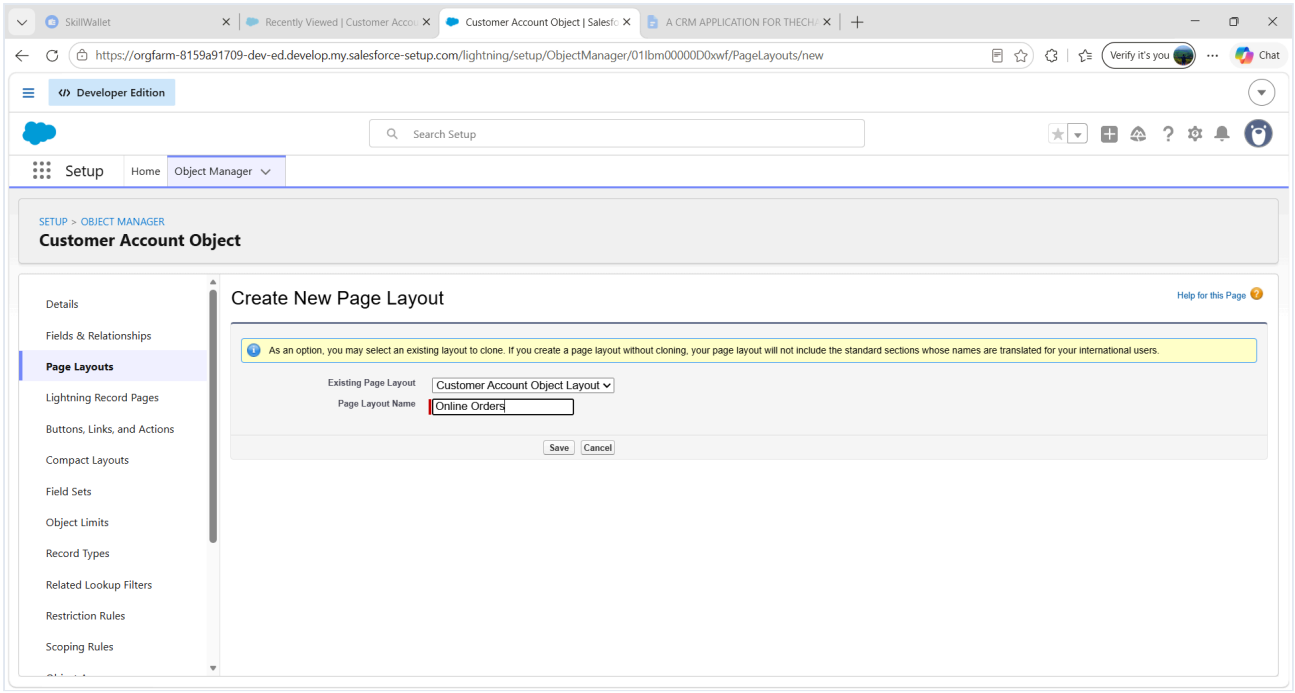


Figure 8: Creating Custom Page Layout for Online Orders

Configuring customized page layouts designed specifically for tea shop online order tracking.

SYSTEM SCREENSHOTS (CONTINUED)

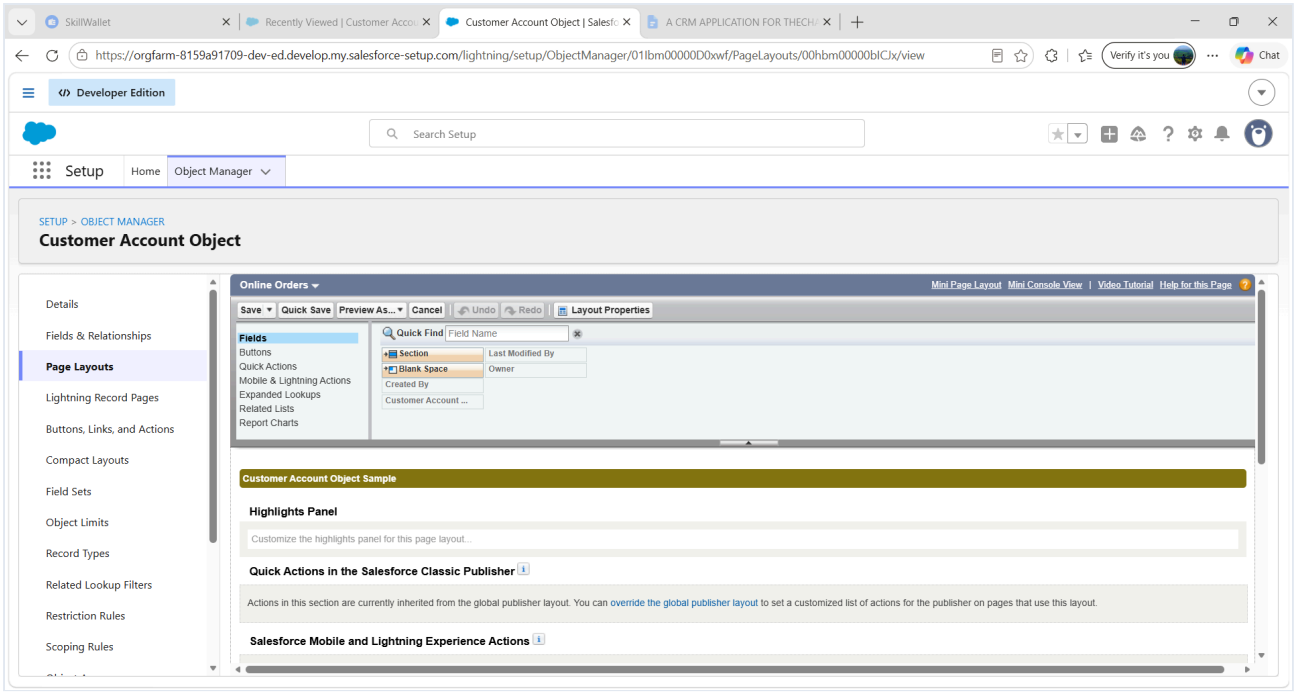


Figure 9: Page Layout Customization Drag-and-Drop Canvas

Customizing highlights panels, quick actions, and field arrangements for customer account objects.

11. TESTING & VERIFICATION

Test Case	Input / Action	Expected Result	Status
TC01	Open application	Dashboard/home page loads	Pass
TC02	Add valid customer	Customer saved in MySQL	Pass
TC03	Submit missing required field	Validation message displayed	Pass
TC04	Edit customer details	Updated values stored correctly	Pass
TC05	Add product catalog entry	Product saved successfully	Pass
TC06	Create customer order	Order items stored correctly	Pass

12. ADVANTAGES & LIMITATIONS

Advantages: Centralized record keeping, reduces paper reliance, simple and responsive interface, expandable architecture.

Limitations: Initial scope covers core CRM routines; advanced payment integrations and live chat require future modules.

13. FUTURE ENHANCEMENTS

- Role-based access control (Admin & Staff logins).
- Customer loyalty and reward points system.
- Automated SMS/WhatsApp order status notifications.
- Integrated billing, invoice generation, and financial reports.

14. CONCLUSION & REFERENCES

The CRM Application for The Tea Shop & Co provides a structured foundation for managing customers, products, and sales orders. By combining modern web interfaces, Python logic, and relational storage, the system minimizes manual paperwork and sets up a robust platform for growth.

References: Python & Flask Documentation, MySQL Reference Manual, MDN Web Docs, Visual Studio Code Guides.